Supporting Information for

Agentic analysis of droplet-engineered appendix-derived gut organoid growth, differentiation, and fusion using AAPS

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This PDF file contains 4 main items including:

Figures S1 to S3

Tables S1

And:

Legends for Source Data-1,2,3,4,5,6

SI References

Other supporting materials for this manuscript include the following:

Source Data-1

Source Data-2

Source Data-3

Source Data-4

Source Data-5

Source Data-6

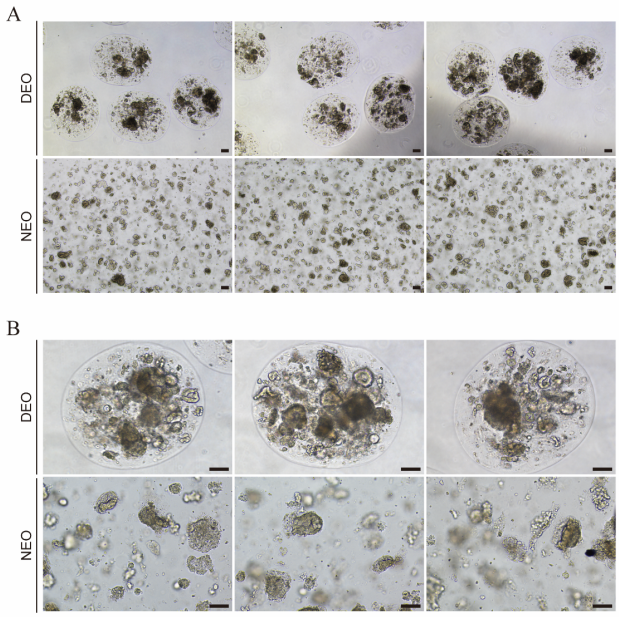


Figure S1 The comparisons between droplet-engineered organoid (DEO) and nonengineered organoid (NEO). Representative brightfield images of Day 0 seeding as crypts in Matrigel domes. Images were captured by using 4× objective lenses (A) and 10× objective lenses (B). Scale bars are 100 μm.

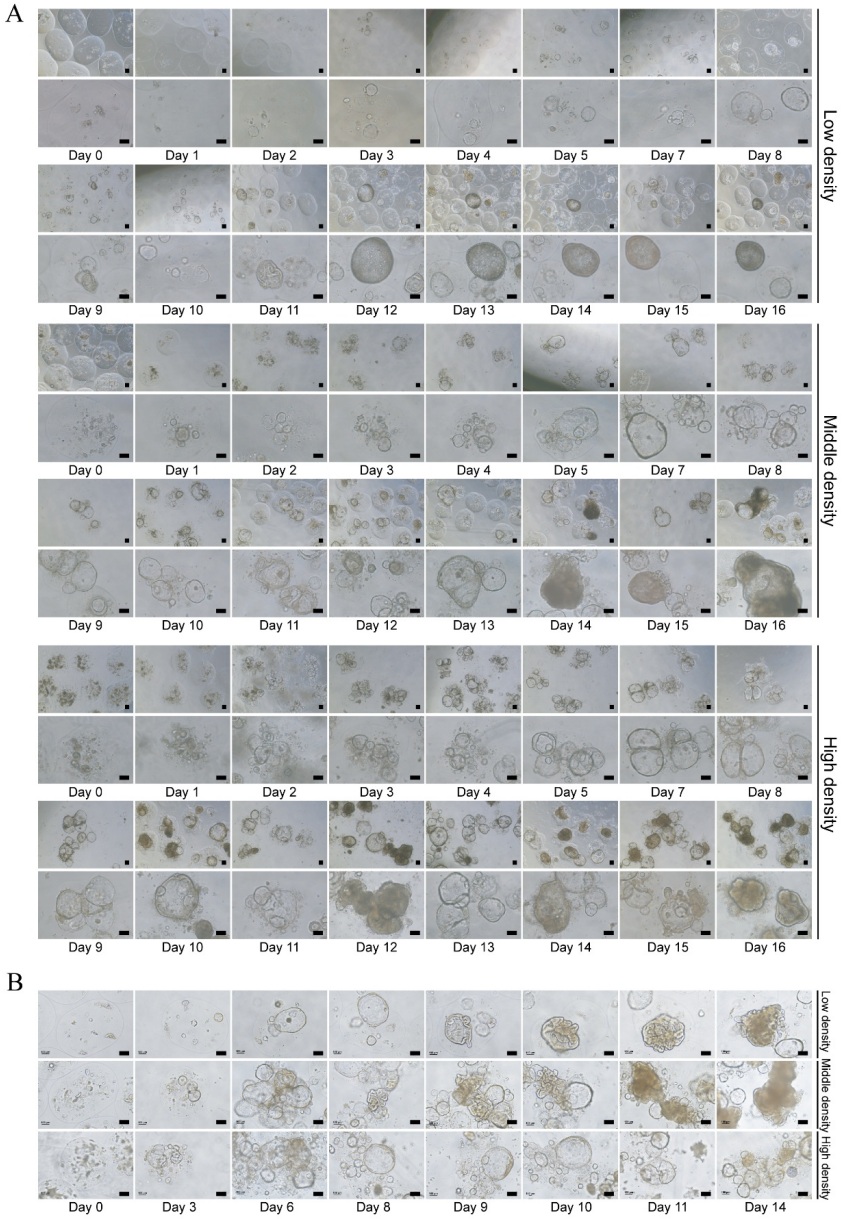


Figure S2 Continued growth of DEOs using low-density, middle-density, and high-density crypt suspensions. Representative images were captured from day 0 to day16. Scale bars are 100 μm.

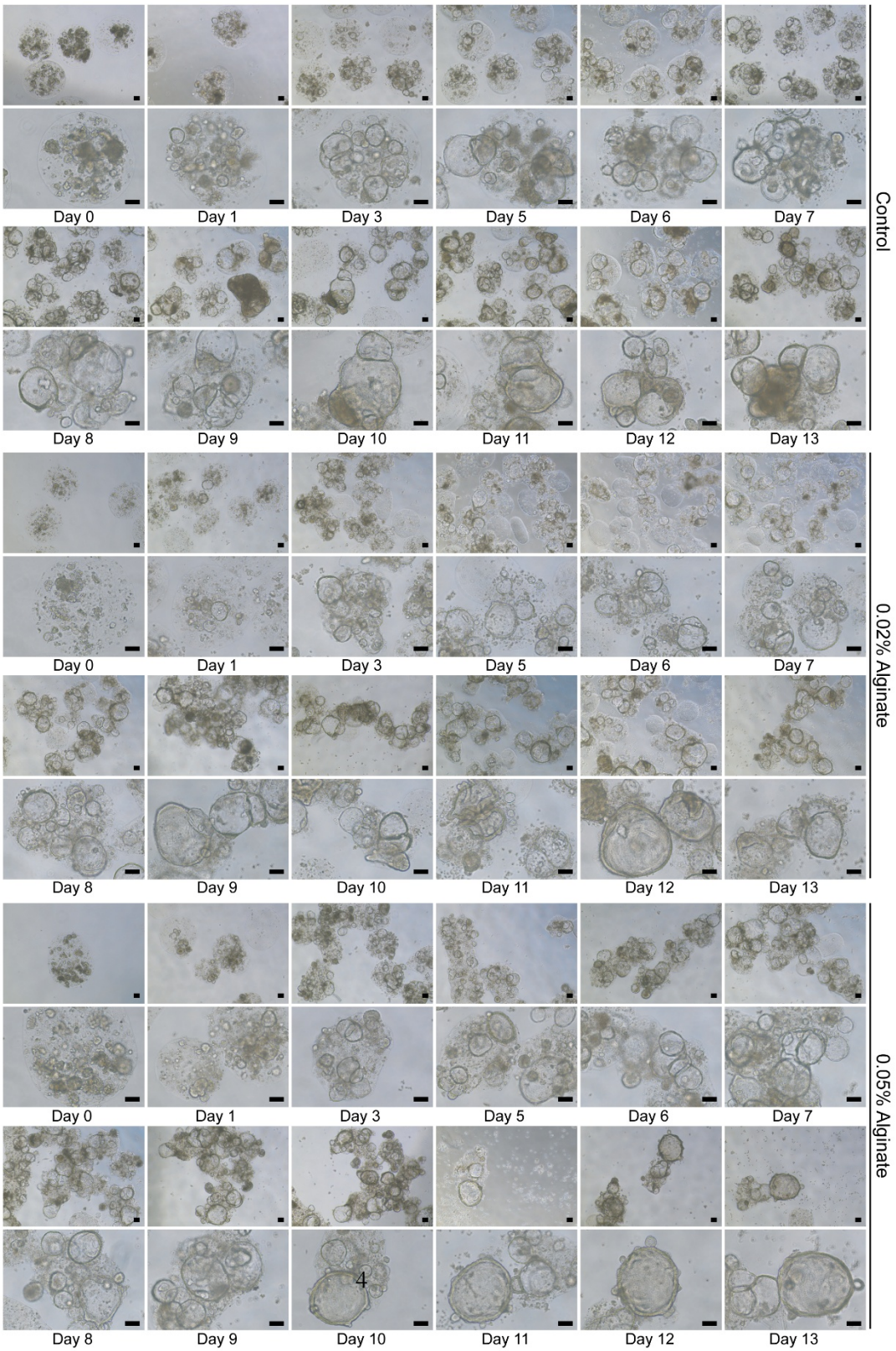
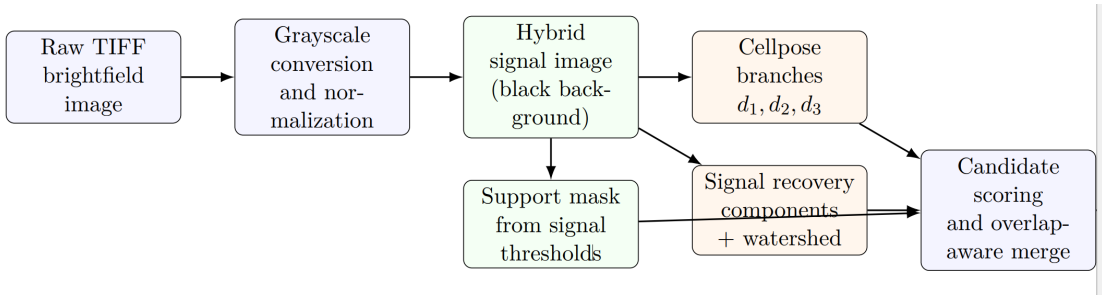

Control
Day 0
Day 1
Day 3
Day 5
Day 6
Day 7
Day 8
Day 9
Day 10
Day 11
Day 12
Day 13
0.02% Alginate
Day 0
Day 1
Day 3
Day 5
Day 6
Day 7
Day 8
Day 9
Day 10
Day 11
Day 12
Day 13
0.05% Alginate
Day 0
Day 1
Day 3
Day 5
Day 6
Day 7
Day 8
Day 9
Day 10
Day 11
Day 12
Day 13

Figure S3 Alginate incorporation of droplet-engineered organoids. DEOs in Matrigel alone (control) or Matrigel adding sodium alginate at final concentrations of 0.02% (low alginate) or 0.05% (high alginate). Representative images were captured from day 0 to day 13. Scale bars are 100 µm.



~~Figure S4 DEO segmentation workflow~~

~~The segmentation workflow generated final instance label maps from brightfield images using hybrid signal preprocessing, multiscale Cellpose branches, and signal-based large-mass recovery, followed by overlap-aware merging of candidate objects~~

（重新做一个好看的放正文）

Figure S4 Representative segmentation outputs for density experiments

Figure S5 Representative segmentation outputs for alginate experiments

Figure S6 Representative segmentation outputs for Y-27632 experiments

Table S1

	Growth metric	Differentiation metric	Fusion metric	Rationale
Density	Total segmented area	Organoid darkness P90	Peripheral-to-central internal edge ratio	Total area is the main segmentation-based growth proxy. Darkness P90 captures darker, more differentiated morphologies. The peripheral-to-central ratio reflects redistribution of edge structure from the organoid periphery toward the interior during fusion.

	Growth metric	Differentiation metric	Fusion metric	Rationale
Alginate	Average perimeter	Wall darkness P90	Peripheral-to-central internal edge ratio	Average perimeter provides a morphology-sensitive readout of organoid expansion. Wall darkness P90 captures darker, thickened epithelial walls associated with differentiation. The same fusion metric was used because it reports the relative shift of edge signal from peripheral regions to the interior.
Y-27632	Mean organoid area (derived from segmented instance areas)	Organoid darkness P90	Center-weighted edge sum	Mean organoid area summarizes organoid size while reducing confounding from fusion-related changes in object number. Darkness P90 captures differentiation-associated darkening. The center-weighted edge sum emphasizes interior edge structure and is therefore sensitive to fusion-associated structural integration.

Abbreviations: P90, 90th percentile; DEO, droplet-engineered organoid.

Source Data 1. Statistical analysis of density experiments at matched time points

Source Data 2. Pairwise statistical comparisons between density groups and control at matched time points

Source Data 3. Statistical analysis of alginate experiments at matched time points

Source Data 4. Pairwise statistical comparisons between alginate groups and control at matched time points

Source Data 5. Statistical analysis of Y-27632 experiments at matched time points

Source Data 6. Pairwise statistical comparisons between Y-27632 groups and control at matched time points

SI References

[1]

[2]